V–M–96d
March 26, 1963

SUPERSEDING
Int. Fed. Spec. V-M-0096c
August 18, 1960, and
Fed. Spec. V-M-96b
December 27, 1956

FEDERAL SPECIFICATION

MATTRESSES, BED: INNER-SPRING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers the construction and materials for inner-spring bed mattresses.

1.2 Classification.

1.2.1 *Types and classes.* The mattresses shall be of the following types and sizes, as specified (6.2):

Type I.—Springs in individual pockets.

Type II.—All-wire spring units, free-end coil construction.

Type III.—All-wire spring units, knotted coil construction.

1.2.1 *Sizes.* Sizes of mattresses shall be as shown in table II, as specified (6.2).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378—Plastic Film (Polyethylene Thin Gage).
T-T-891—Twine, Linen.
V-T-276—Thread, Cotton.
CCC-C-430—Cloth, Cotton, Sheeting.
CCC-C-436—Cloth, Cotton, Ticking, Twill.
CCC-T-191—Textile Test Methods.
DDD-L-20—Label; For Clothing, Equipage, and Tentage, (General Use).
DDD-T-86—Tape, Textile; Cotton, General Purpose (Unbleached, Bleached or Dyed).
PPP-B-576—Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-636—Box, Fiberboard.
PPP-B-640—Boxes, Fiberboard, Corrugated, Triple Wall.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 751—Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta,

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Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129—Marking for Shipment and Storage

MS-16686—Medical Services Symbol

(Copies of Military Specifications and Standards required by the contractors in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

U. S. Department of Agriculture Publication:

SRA-180—Cotton and Cotton Linter Standards.

(Application for copies should be addressed to the Department of Agriculture, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Felt.* The felt shall be cleaned and shall be well garnetted to produce unbroken laminated sheets equal to a standard sample (6.4) made of a blend of 50 percent cotton conforming to 3.1.1.1 and 50 percent linters conforming to 3.1.1.2. The felt for padding the border ticking in the border construction shall weigh not less than 1.72 ounces per linear yard. The felt padding for reinforcing both edges of the border shall weigh not less than 1 ounce per linear yard (3.3.1.3.1). The felt padding used in alternate border construction (3.3.1.3.2) shall weigh not less than 2 ounces per linear yard.

3.1.1.1 *Cotton.* The cotton shall be new (not previously used), 7/8-inch staple cotton, not lower in grade than middling, conforming to the standard of the United States Department of Agriculture Publication No. SRA-180 (2.2).

3.1.1.2 *Linters.* The linters shall be new (not previously used), not lower in grade than United States standard grade No. 3 and staple 3, conforming to the United States Department of Agriculture Publication No. SRA-180 (2.2).

3.1.2 *Ticking.* The ticking shall conform to the requirements of type I ticking of CCC-C-436.

3.1.3 *Sheeting.* The cotton sheeting shall conform to type I, class 1, style A of CCC-C-430.

3.1.4 *Thread, sewing.* The cotton thread shall conform to type IA1 or IA2, ticket No. 30, 3-ply of V-T-276.

3.1.5 *Tufting twine, braid, and tape.* The twine or braid used in tufting shall have a breaking strength of not less than 32 pounds. Tufting tape shall be not more than 5/8 inch wide and shall have a breaking strength of not less than 50 pounds. Testing shall be as specified in 4.2.1.

3.1.6 *Twine, stitching.* The twine for stitching mattresses shall conform to type II of T-T-891.

3.1.7 *Tape, cotton.* The tape used in finishing the mattress edges (3.3.1.6 and 3.3.2.6) shall be 5/8-inch (minimum) cotton tape conforming to class 1 or 2 of DDD-T-86.

3.1.8 *Tape with metal locking strip.* The tape shall be single 5/8-inch cotton tape conforming to class 1 or 2 of DDD-T-86. The metal locking strip shall be made from a

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corrosion-resistant steel, of a type commercially available and commonly used in tufting mattresses.

3.1.9 *Hog rings and wire clips.* The hog rings (3.3.2.2) shall be commercial type, of the size, strength, and finish commonly used in mattress construction. The wire clips for securing ends of rope lifting straps (3.6.2) shall be a commercial type.

3.1.10 *Rope for lifting strap.* When rope is used in making alternate lifting straps (3.6.2) it shall be a commercial type 3/8-inch natural cotton rope.

3.1.11 *Coils.*

3.1.11.1 *Coils for type I mattress.*

3.1.11.1.1 *Coil wire.* The wire for the spring coils shall be not less than 16-gage wire (nominal 0.0625-inch diameter). The wire shall be hard-drawn, unholstery-grade, spring-steel wire with a minimum tensile strength of 240,000 p.s.i. and a maximum tensile strength of 280,000 p.s.i., when tested as specified in 4.2.1. The wire shall have a protective zinc coating which will not be visibly affected by salt spray, nor will there be visible evidence of rust when tested as specified in 4.2.1.

3.1.11.1.2 *Coil design.* The coils shall be approximately 2-7/16 inches in diameter, cylindrical or barrel shaped, and shall have both ends of wire turned in. There shall be not less than 3 nor more than 3-1/2 turns per coil, measured on the free coil, not including end bearing convolutions. Coils shall be given suitable thermal treatment to relieve residual stresses caused by coiling. The finished coils shall not decrease in height more than 1/16 inch when tested as specified in 4.2.1.

3.1.11.2 *Coils for type II mattress.*

3.1.11.2.1 *Coil wire.* The wire for the edge and corner coils shall be not less than 13-gage wire (nominal 0.0915-inch diameter). The wire shall be hard-drawn, upholstery-grade, spring-steel wire with a minimum tensile strength of 205,000 p.s.i., and a maximum tensile strength of 250,000 p.s.i., when tested as specified in 4.2.1. The wire for all other coils shall be not less than 13-1/2-gage wire (nominal 0.086-inch diameter). The wire shall be hard-drawn, upholstery-grade, spring-steel wire with a minimum tensile strength of 212,500 p.s.i., and a maximum tensile strength of 275,500 p.s.i., when tested as specified in 4.2.1.

3.1.11.2.2 *Coil design.* Corner coils shall be cylindrical, with not less than 6 convolutions. The ends of the coils shall be die-formed to a 4-inch radius. End coils shall be offset, cylindrical or barrel overlapping-type coils. Edge coils forming the sides of the spring unit shall present a straight edge to the outer edge of the spring unit and shall be cylindrical. All other coils shall be hourglass shape. All coils except the corner coils shall have not less than 4-1/2 complete convolutions. All coils shall be free-end type coils to allow their being anchored within the helical without accessory locking device. The coils shall be 5 inches ± 1/4 inch in height. Coils shall be given suitable thermal treatment to relieve residual stresses caused by coiling. Coils shall not decrease in height more than 3/16 inch when tested as specified in 4.2.1.

3.1.11.3 *Coils for type III mattress.*

3.1.11.3.1 *Coil wire.* The wire for the spring coils shall be not less than 13-1/2-gage wire (nominal 0.086-inch diameter). The wire shall be hard-drawn, upholstery-grade, spring-steel wire with a minimum tensile strength of 212,500 p.s.i., and a maximum tensile strength of 275,500 p.s.i., when tested as specified in 4.2.1.

3.1.11.3.2 *Coil design.* All coils shall be hourglass in shape, except that the coils in the two longitudinal rows forming the sides of the spring unit may be cylindrical or elliptical in shape, provided they conform

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to good industry practice. There shall be not less than three active turns or convolutions per coil, including only portions between the knots of the coil, and the coils shall be 4-3/4 inches ± 1/4 inch in height. Coils shall be given suitable thermal treatment to relieve stresses caused by coiling. Coils shall not decrease in height more than 3/16 inch when tested as specified in 4.2.1.

3.1.12 *Helicals for types II and III mattress.* The helical wire shall be made from hard-drawn high-carbon steel wire, not less than 17-gage (nominal 0.054-inch diameter).

3.1.13 *Ventilators.* The ventilators shall be of one piece, of brass or aluminum not less than 0.013 inch thick, and with an outside diameter of 7/8 to 1 inch. The length of the ventilator shank shall be 5/16 inch, with a diameter of 11/16 to 3/4 inch. At least 61 percent of the face of the ventilator shall be perforated with not less than 61 holes 1/32 to 1/16 inch in diameter.

3.1.14 *Tufting buttons.* The tufting buttons shall be of commercial-type fiber-plastic material or cold-rolled steel, with a shank made of wire with a diameter of not less than 3/64 inch nor more than 4/64 inch, and the shank shall be of such shape that its top shall form a shoulder. The shank shall be secured and shall be centered in the button, and the outer dimension from shoulder tip to shoulder tip shall be not less than 17/64 inch nor more than 20/64 inch. The space between the top of the shoulder and the underside of the button shall be not less than 4/64 inch nor more than 5/64 inch. The top of the button shall be flat, convex, or concave and shall have a diameter of not less than 1/2 inch nor more than 5/8 inch. When metal tufting buttons are used, they shall be given a protective coating or plating which will not be visibly affected by salt spray nor will there be visible evidence of rust when tested as specified in 4.3.3.2.

3.1.15 *Stabilizer wire (type II only).* The stabilizer wire shall be made from upholstery, high-carbon, oil-tempered wire, not less than 15-gage (0.072-inch diameter), with corners die-formed to a 4-inch radius.

3.1.16 *Wire mat.* When wire mat is used as the spring covering specified in 3.3.2.2, it shall conform to the following: The wire mat shall have a series of steel wires uniformly spaced not more than 1 inch apart, running at right angles to longitudinal fiber cords which are uniformly spaced not more than 8 inches apart. The wires shall pierce and pass through the center cords and shall be crimped around the border cords. The wires shall be 19-gage (nominal 0.041 inch in diameter) oil-tempered steel wire with a tensile strength range of 266,000 p.s.i. to 296,000 p.s.i. The spirally wound fiber cords shall be constructed of not less than 30-pound kraft paper. The center cords shall be not less than 0.16 inch in diameter, with or without a three-strand sisal or jute core, and shall have a breaking strength of not less than 100 pounds after being pierced. The border cords shall be not less than 0.125 inch in diameter with a breaking strength of not less than 100 pounds. Testing shall be as specified in 4.2.1.

3.2 Stitches and seams.

3.2.1 *Stitches.* Stitching shall conform to stitch type 301 or 401 of Fed. Std. No. 751 with 8 to 12 stitches per inch.

3.2.2 *Seams.* When longitudinal pieces of ticking are used (3.3.1.5), they shall be stitched with stitch type 401 and seamed with seam type LSb-2 or stitch type 301 and seam type SSa-1 conforming to Fed. Std. No. 751.

3.2.3 *Thread breaks and ends of seams.* Ends of all seams and stitching produced with 301 stitch type when not caught in other seams or stitching shall be backtacked not less than 1/4 inch. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks (401

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stitch type) may be repaired using 301 stitch type.

3.3 Construction.

3.3.1 *Type I mattresses (springs in individual pockets).*

3.3.1.1 *Inner-spring construction.* The coils specified in 3.1.11.1 shall be contained in a continuous string of sheeting pockets made from sheeting specified in 3.1.3, each pocket containing one coil. The coils shall stand upright and shall be parallel to one another. The coils shall be not less than 4 inches in height when confined only by the pockets. The side stitching between individual pockets for each coil shall be done with one row of machine stitching extending from top to bottom of pocket. The top of pockets shall be seamed closed by machine. The pockets shall be such as to accommodate the coil, leaving only sufficient slack to permit proper spring action. Each coil shall be held in the sheeting pocket at all times in a manner to insure positive pressure against both the top and bottom layers of felt in the mattress, so as to provide against the development of slackness or looseness between the spring and the top and bottom felt layers. The inner-spring assembly shall be built by alternately weaving the continuous rows of sheeting bagged springs lengthwise. Each row of coils shall nest between the adjacent row in a manner to allow overlapping of the coils, and provide against any open spaces between the sheeting pocketed coils. After being enclosed in pockets and laid up in mattress formation, the individual spring coils shall be tied together at the ends through the pockets in the direction of the width of the assembly with rows of twine spaced not more than 9 inches apart in the direction of the length. The twine shall be as specified in 3.1.6.

3.3.1.2 *Fitted sheeting cover construction.* The upper and lower surfaces of the inner-spring assembly shall be protected by sheeting covers made from sheeting specified in 3.1.3. Covers are to overlap all edges of inner-spring construction and shall be attached to spring construction and assembled border all around by twine stitching. Stitches shall be spaced to engage and secure each border coil at sides and every other coil across ends of sheeting cover and assembled border; each stitch to be suitably knotted.

3.3.1.3 *Border construction.* The border shall be constructed in accordance with 3.3.1.3.1 or 3.3.1.3.2.

3.3.1.3.1 *Border construction.* The outside of the border on the mattress shall be made of ticking specified in 3.1.2 with both edges hemmed with machine stitching with a minimum 1/4-inch hem. The border on each mattress shall be made of one piece all around, with both ends hemmed, spliced, and securely sewn together by machine so as to leave no break in its uniformity and finish. The border ticking shall be padded with sufficient layers of felt specified in 3.1.1. The felt padding shall be covered with sheeting. The felt and sheeting shall be secured at the border with transverse stitching from top hem to bottom hem 1-1/2 inches plus or minus 1/4 inch on centers. Both edges of the border shall be reinforced with additional felt specified in 3.1.1, approximately 2-3/8 inches wide, backed by sheeting not less than 1 inch wide, by stitching to the border through the felt and ticking at the transverse stitching. The border shall have ventilators specified in 3.1.13 and table II, placed in a line half way between the top and bottom hem of the border and equidistant on the sides and ends. Ventilators shall pass through the entire border from the outside through to the spring section. The ventilators shall securely engage the inner and outer edges of the round openings punched through all layers of the border including the felt, ticking, and sheeting. The ventilators shall be of a type, the entire border so constructed, and the felt in the border shall be so placed, as not to interfere with the free passage of air through the ventilators into the spring section, thus admitting proper ventilation to the inside

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of the mattress. Details necessary for proper construction of the border not specified herein are left to the option of the contractor.

3.3.1.3.2 *Alternate border construction.* The outside of the border shall be made of ticking specified in 3.1.2 with both edges hemmed with a minimum 1/4-inch hem. The border shall be made of one piece of ticking with both ends hemmed, spliced, and sewn together by machine in a continuous manner so as to leave no break in its uniformity and finish. The ticking shall be padded with additional layers of felt as specified in 3.1.1, on the basis of a 5-inch side border, and shall be at least the width of the hemmed border plus 1/4 inch or minus 1/2 inch. The felting shall be butted or spliced at the seam joining the ends of the border in such a manner as to leave no break in its uniformity and finish. Felt shall be covered with sheeting specified in 3.1.3 of a width at least equal to the width of the border. The ticking, felt, and sheeting shall be sewn together with a commercial pattern of not less than six stitches per inch. The border shall be equipped with ventilators as specified in 3.3.1.3.1.

3.3.1.4 *Padding springs.* The springs and covering material shall be padded with cotton felt, specified in 3.1.1, placed over the sheeting cover in full length and full width layers. From 85 to 95 percent of the cotton felt specified in table II shall be distributed in layers of equal weight on both faces of the mattress. These layers shall produce uniform thickness with the exception that the remaining 5 to 15 percent of the total weight of felt on each face shall be uniformly distributed over the middle one-third of the length of the face for the full width of the mattress. Both sides of the mattress shall be finished so that either side may be used as the sleeping side with equal comfort.

3.3.1.5 *Cover construction.* The top and bottom sections in the mattress cover shall each be made up of not more than two longitudinal pieces of ticking specified in 3.1.2. The ticking shall be matched and sewn together (3.2.2). The blue stripe of the ticking shall run in the lengthwise direction of the mattress.

3.3.1.6 *Edge construction.* The border at the edge of the spring assembly shall be padded with additional layers of cotton felt, so held in place as to give a box effect to the outside edge of the mattress. The construction along the inner edge shall be suitably reinforced and finished to maintain a sharply defined box edge. All spring coils adjacent to the prebuilt border of the box edge shall have the top and bottom of the coil sewn to the border with twine specified in 3.1.6 except that when borders are constructed as specified in 3.3.1.3.2, the four spring coils on either side of each corner need not be stitched to the edge of the border. Additional felt shall be used to form a well-rounded corner. The hemmed edges of the cover ticking shall be sewn to the edges of the border ticking and the edge of cover and border shall be finished and full sewn within the folds of a tape as specified in 3.1.7. This sewing shall be done by machine and shall extend completely around the edges of the mattress.

3.3.2 *Type II mattresses (all-wire spring unit, free-end coil construction).*

3.3.2.1 *Spring unit.* The spring unit shall be made with materials specified in 3.1.11, 3.1.12, and 3.1.15 located so as to keep edges from sagging, prevent side sway, to give firm support to all parts of the body, and to maintain flexibility. All-wire spring units, free-end coil construction, shall consist of transverse rows of alternating right-hand and left-hand coils connected with a helical running at right angles to the long axis of the unit in such a manner as to prevent lateral slippage. The free end of the coil shall be anchored within the helical without the use of any accessory locking device. The edges of all outside coils shall be constructed so as to present a straight

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edge to the construction. The individual coils forming the end rows shall have a double overlap of the end convolutions into the end helical and second helical. The end rows shall have a stabilizer wire specified in 3.1.15 placed within each of the 4 end helicals with the free ends of the wire extending to a point on line with the center of the second row of coils at each end, and fastened so as to prevent slippage. The 4 free-end corner coils of the unit construction shall be cylindrical. Helicals connecting the transverse rows of coils shall be not over 0.321-inch outside diameter. The helicals connecting the end coils shall be not over 0.350-inch outside diameter. All helicals shall have a 3/8-inch pitch or less. The ends of the helicals shall be turned in to form a smooth, rounded termination. The spring unit shall be finished with a baked-on enamel, or rust-resisting metallic coating.

3.3.2.2 *Spring covering.* The spring unit shall be covered with sheeting as specified in 3.3.1.2, or wire mat specified in 3.1.16. The wire mat shall be attached to both sides of the inner-spring unit fastened to the long edges of the unit by means of hog rings specified in 3.1.9. The rings shall be clamped around one wire of the mat and the top of an edge coil of the inner-spring unit. On one side of the spring unit, the wire mat shall be attached at one edge with three hog rings, two at the ends and one in the middle. The mat shall be fastened on the opposite edge with two hog rings each one-quarter length from the ends.

3.3.2.3 *Border construction.* The border shall be constructed as specified in 3.3.1.3.1 or 3.3.1.3.2.

3.3.2.4 *Padding springs.* The cotton felt padding shall be applied over the springs as specified in 3.3.1.4.

3.3.2.5 *Cover construction.* The cover shall be constructed as specified in 3.3.1.5.

3.3.2.6 *Edge construction.* Mattresses shall have a prebuilt border as specified in 3.3.1.3.2. The border at the edge of the spring assembly shall be padded with cotton felt so as to give the outside edge of the mattress a box effect. The construction along the inner edges shall be suitably reinforced and finished to maintain a sharply defined box edge. The hemmed edge of the cover ticking and the hemmed edge of the border shall be joined together and sewn within the folds of a white tape as specified in 3.1.7. This shall be done by machine sewing, extended completely around the edge of the mattress. Not more than two pieces of tape shall be used on each border edge.

3.3.3 *Type III mattresses (all-wire spring unit, knotted-coil construction).* Type III mattresses shall be constructed in accordance with the requirements for type II as specified in 3.3.2 except the requirements for spring unit shall be as specified in 3.3.3.1.

3.3.3.1 *Spring unit.* All-wire spring units, knotted-coil construction, shall consist of transverse rows of coils specified in 3.1.11.3, connected by helicals running at right angles to the long axis of the unit to prevent lateral slippage and to permit a free-hinge action when the mattress is folded or rolled from end to end, or shall consist of transverse rows of individual coils or units of not more than 4 coils to a unit, joined to the neighboring coils or units of coils to prevent lateral slippage and to permit a free-hinge action when the mattress is folded from end to end. Helicals, when used, shall be 5/16- (plus or minus 1/16) inch outside diameter, and shall have not less than twenty-four turns per foot. Where helicals are used in the construction, the individual spring coils may be offset with a portion or portions passing through not less than three turns of the helical. All knots shall be turned down and the ends of all helicals shall be turned inward forming a smooth and rounded termination. The unit shall be finished with a baked-on enamel, or rust-resisting metallic coating.

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TABLE I.- Tufting requirements

Size	1	2	3	4	5	6	7	8	9	10	11	12	13
Number of tufts 1/	17	28	28	28	28	28	28	28	39	39	39	39	28
Pattern 2/	2x6	3x6	3x6	3x6	3x6	3x6	3x6	3x6	4x6	4x6	4x6	4x6	3x6

1/ The spacing of the tufts shall be uniform. For mattresses of 80 inches or longer, there shall be an additional tuft in each lengthwise row for every 10 inch increase in length.

2/ The arrangement of the tufts shall be in a diamond pattern. The distance of any row closest to an edge shall be 6 inches plus or minus 1 inch.

3.4 Tufting. Mattresses shall be tufted on both sides using either button tufts specified in 3.1.14, tufting twine, braid, or tape, specified in 3.1.5 or tape with a corrosion-resistant metal locking strip, specified in 3.1.8, fastened to each end. Knots of tape tying tufts shall be arranged so that they will not appear above the surface of either side of the mattress. Button tufts shall be drawn on both sides to a length not more than the thickness of the mattress, minus 2 inches. The number of tufts for each size mattress shall be as specified in table I.

3.5 Corners. The four corners of each mattress shall be rounded on a 4-1/2-± 1/2-inch radius.

3.6 Lifting straps.

3.6.1 *Vertical lifting straps.* Unless otherwise specified (6.2), two vertical lifting straps of double-thickness ticking, with turned-in sewed edges, shall be attached to each side of the mattress, at approximately one-third of the length from each end. Each strap shall measure 1-3/4 inches ± 1/8 inch wide, and of a length consistent with good commercial practice. Both ends of the strap shall be attached to the mattress border with 4 rows of 301 stitching, centered on each end with the ends of stitching backstitched 3/8 to 1/2 inch. Each row of stitching shall be 1-1/2 inches ± 1/16 inch long and stitching shall not go off the strap. Each row shall be spaced 1/16 to 1/8 inch apart with the first row 1/8 to 3/16 inch from raw edge of strap. The rows of stitching shall not be cross stitched. Or, both strap ends shall be attached to the mattress border with not less than two bartacks centered on each end, each 1-1/2 inches long, with 42 stitches per bartack. Each bartack shall be spaced 1/8 to 3/16 inch apart with the first bartack 1/8 to 3/16 inch from the raw edge of strap. Or, both strap ends shall be attached to the mattress border with not less than two linetacks centered on each end, each 1-1/2 inches long, with 30 to 34 stitches per linetack. Each linetack shall be spaced 1/8 to 3/16 inch apart with the first linetack 1/8 to 3/16 inch from the raw edge of strap. In addition, the raw ends of the strap shall be caught under the cotton tape, binding the edges of the mattress.

3.6.2 *Alternate lifting straps.* When specified (6.2), two horizontal lifting handles shall be attached to the border on each side of the mattress. The handles shall be either made of cotton rope as specified in 3.1.10, passed through 1/2-inch nonferrous grommets, securely knotted behind the grommets and with the ends of the rope secured together with wire clips (3.1.9), or a commercial, braid-covered, cotton-cord lifting strap utilizing nonferrous eyelets or grommets, metal clip cord ends and a metal back-

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TABLE II.- Physical requirements

Sizes	1	2	3[1/]	4	5	6	7	8	9	10	11	12	13[2/]
Length (inches), plus or minus 1/2 inch	76	78	76	75	78	84	75	78	78	78	73	75	80
Width (inches), plus or minus 1/2 inch	29	33	34	36	36	36	38	39	41½	47	53	53	36
Thickness (inches), plus or minus 1/4 inch	6½	6½	6½	6½	6½	6½	6½	6½	6½	6½	6½	6½	6½
Weight of felt layer on each face (min.)(lbs.)	8½	10	8¼	10½	11	12	11½	12	13	15	16	16	11
Number of ventilating openings:													
In each side	4	4	4	4	4	4	4	4	4	4	4	4	4
In each end	2	2	2	2	2	2	3	3	3	3	3	3	2

Type I mattresses (springs in individual pockets)

Sizes	1	2	3[1/]	4	5	6	7	8	9	10	11	12	13[2/]
Weight (min.)(lbs.)	39	47	48	53	55	55	58	55	61	70	70	78	2/
Weight (max.)(lbs.)	45	54	56	61	63	63	65	63	68	77	79	88	2/
Number of coils in row:													
In direction of length (minimum)	31	32	31	31	32	33	31	32	32	32	30	31	2/
In direction of width (minimum)	15	17	18	18	18	18	19	19	20	22	24	24	2/

Type II mattresses (all-wire spring unit, free-end coil construction)

Sizes	1	2	3[1/]	4	5	6	7	8	9	10	11	12	13[2/]
Weight (min.)(lbs.)	30	39	1/	38	41	47	45	47	50	57	58	60	44
Weight (max.)(lbs.)	35	44	1/	43	45	53	49	53	57	64	65	67	50
Total number of coils (minimum)	126	152	1/	162	171	180	162	171	190	209	221	234	171

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TABLE II. (*cont'd*)

Type III mattresses (all-wire spring unit, knotted-coil construction)													
Sizes	1	2	3 1/	4	5	6	7	8	9	10	11	12	13 2/
Weight (min.)(lbs.)	30	30	1/	36	39	41	43	41	43	50	51	60	36
Weight (max.)(lbs.)	35	34	1/	41	43	47	47	47	49	57	58	67	40
Total number of coils (minimum)	126	157	1/	162	171	180	162	171	190	209	221	234	171

1/ Size 3 is not applicable to types II and III.
2/ Size 13 is not applicable to type I.

ing strip for each handle. The finished length of the handles and spacing of the grommets or eyelets shall be consistent with good commercial practice.

3.7 Deviation of mattress surface (type I only). The maximum average deviation of mattress surface shall be not greater than 3 degrees and the maximum deviation of mattress surface at any place shall be not greater than 6 degrees, when tested as specified in 4.2.4.2.

3.8 Depression under compression-plunger (type I only). The difference between the minimum and maximum depression produced by the plunger across the mattress under loads of 20 pounds shall be not greater than 25 percent of the height of the mattress, when tested as specified in 4.2.4.2. The depression at any point shall be not greater than 60 percent of the height of the mattress.

3.9 Physical requirements. Mattresses shall conform to the requirements shown in table II.

3.10 Labels. All labels shall be sewn into the same edge seam at either end of each mattress.

3.10.1 *Statutory labels.* Statutory labels shall conform to the legal requirements of the state in which the mattresses are manufactured.

3.10.2 *Identification labels.* Each mattress shall have an identification label conforming to type I, class 5 of DDD-L-20.

3.10.3 *Medical label.* When specified (6.2), mattresses procured for medical activities shall have a label conforming to type I, class 9 of DDD-L-20, bearing manufacturer's name or registered trademark and a caduceus approximately 1 inch in height centered between "U" and "S" of appropriate size conforming to type I of MS-16686.

3.11 Workmanship. The finished mattresses shall conform to the quality and grade of product established by this specification. Occurence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Inspection for acceptance. Except when otherwise specified by Civilian Government Activities, or where otherwise specified herein, inspection shall be in accordance with the provisions of MIL-STD-105.

4.2.1 *Inspection of components.*

4.2.1.1 *Testing of components.* In addition to the quality assurance testing provisions of the subsidiary specifications and drawings, testing shall be performed on components and materials listed in table III for the characteristics shown therein. The inspection level (sample size) shall be determined from MIL-STD-105, utilizing Inspection Level L-1, AQL 6.5 DHU, for sample unit requirements, and Inspection Level L-1 for lot average requirements. When the data in the "number of determinations per sample unit" and the "results reported as" columns are not specified in the following table, they shall be as required by the referenced test method.

4.2.1.2 *Certification of compliance.* Component(s) and material(s) listed below may be accepted on the basis of a contractor's certification of compliance for requirements specified in applicable paragraphs of this specification.

Component	Characteristic	Requirement paragraph
Coils	Finish Diameter of wire Diameter of coil Shape of coil Style of coil Number of turns per coil	3.1.11.1, 3.1.11.2, and 3.1.11.3
Ventilators	Construction Material identification Thickness Overall outside diameter Length of shank Diameter of shank Percent of face that is perforated	3.1.13
Ventilators (cont'd)	Number of holes Diameter of holes	3.1.13
Tufting buttons	Material identification Construction Diameter of shank wire Shape of shank Positioning and security of shank Outer dimension (shoulder tip to shoulder tip) Distance between top of shoulder and underside of button Finish on entire metal shank and metal button	3.1.14
Metal locking strip	Corrosion - resistant steel	3.1.8
Felt (layers in border)	Weight (1.72 oz./linear yard) Weight (1.0 oz./linear yard) Weight (2.0 oz./linear yard)	3.1.1, 3.3.1.3.1, and 3.3.1.3.2
Felt layers on each face	Weight	Table II
Spring unit (Types II and III mattresses	Finish and treatment	3.3.2.1
Cotton rope (for lifting straps)	Material identification Diameter	3.1.10
Hog rings	Commercial type	3.1.9
Wire clips	Commercial type	3.1.9

4.2.2 *In-process inspection.* Inspection may be made at any point or during any phase of the manufacturing process to determine whether operation or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated non-conformance. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specification requirements.

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Operation or assembly	Requirement paragraph
Construction of wire mat for spring covering (when applicable) - types II and III mattresses	3.1.15
Construction of inner-spring assembly, type I mattress	3.3.1.1
Construction of inner-spring assembly, type II mattress	3.3.2.1
Construction of inner-spring assembly, type III mattress	3.3.3.1
Application of cotton felt (types I, II, and III mattresses)	3.3.1.4
Fabrication of border	3.3.1.3.1 or 3.3.1.3.2
Edge construction (type I mattress) (type II and III mattresses)	3.3.1.6 3.3.2.6
Fabrication of alternate lifting straps	3.6.2
Attachment of wire mat or sheeting (when applicable) for spring covering	3.3.1.2 or 3.3.2.2

4.2.3 *Intermediate inspection.* The defects found during examination shall be in accordance with 4.2.3.1, 4.2.3.2, 4.2.3.3, and 4.2.3.4. The lots shall be inspected in accordance with 4.2.3.5.

4.2.3.1 *Examination of inner-spring assembly (type I mattress) for visual defects.* The sample unit for this examination shall be one inner-spring assembly protected by a fitted sheeting cover. Examination of inner-spring assembly for defects in material and workmanship.

Examine	Defect
Sheeting	Any material defect such as a hole, smash, float, or slub which may develop into a hole.
	Any workmanship damage such as a cut, tear, rip, mend, burn, or needle chew which may develop into a hole.
Construction and workmanship	Both upper and lower surfaces of assembly not covered by a sheeting cover.
	Sheeting cover does not overlap all edges of inner-spring construction.
	Sheeting cover not attached to spring construction and assembled border all around by twine stitching.
	Stitches do not engage and secure each border coil at sides and every other coil across ends to sheeting cover and assembled borders.
	Any stitch that is not knotted.
	More than two stitches not adequately knotted.

4.2.3.2 *Examination of the all-wire spring unit, free-end coil construction assembly (type II mattress) for visual defects.* The sample unit for this examination shall be one type II mattress.

Examine	Defect
Construction	Transverse rows do not consist of alternating right-hand and left-hand coils.
	Alternating right-hand and left-hand coils are not connected with a helical running at right angles to the long axis of the unit in such a manner as to prevent lateral slippage.
	Free end of coil is not anchored within the helical.
	Intermediate convolutions of the individual coils in the longitudinal rows forming the side are not cylindrical in shape.
	Edges of all outside coils are not constructed so as to present a straight edge to the construction.
	The individual coils forming the end rows are not offset cylindrical or barrel type and are not overlapped to form closed end or bearing convolutions.
	Corners of stabilizer wires are formed to less than 3-1/2-inch or more than 4-1/2-inch radius.
	The free end of the stabilizer wire is not placed within each of the 4 end helicals and the free ends of the wire do not extend to a point or line with the center of the second row of coils at each end and not fastened so as to prevent slippage.
	Less than 6 convolutions on any free-end corner coil.
	Any free-end corner coil not cylindrical in shape.
	Any corner coil formed to less than 3-1/2-inch or more than 4-1/2-inch radius.
	Not an hourglass shape on any other coil not indicated above.
	Coil not indicated above having less than 4-1/2 complete convolutions (when counted from last point of contact of coil with helical on top to the first point of contact with the helical on the bottom).
	Coil having a height less than 4-3/4 inches or more than 5-1/4 inches.
	Pitch of helicals is more than 3/8 inch.
	End of helicals not turned in forming a smooth rounded termination.

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TABLE III.- Test methods of components

Component and lot expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	Number determinations per sample unit	Results reported as	Sample unit
Garnetted felt (100 lbs)	Quality	3.1.1	Visual	-	X	1	Pass or fail	12 inches by 12 inches
Cotton (1 bale)	Staple length	3.1.1.1	Visual	-	X	1	Pass or fail	1/2 pound
	Grade	3.1.1.1	Visual	-	X	1	Pass or fail	1/2 pound
Linters (1 bale)	Grade and staple	3.1.1.2	Visual	-	X	1	Pass or fail	1/2 pound
Tufting twine or braid (1 cone or spool)	Breaking strength	3.1.5	4102, CCC-T-191	X	-	-	-	1 cone or spool
Tufting tape (1 spool)	Breaking strength	3.1.5	4.3.4.1	X	-	-	-	1 cone or spool
	Width	3.1.5	Standard Commercial	-	X	3	Nearest 1/16 in.	1 cone or spool
Coil	Material identification	3.1.11	Standard Commercial	X	-	1	Pass or fail	
	Tensile strength	3.1.11	Standard Commercial	-	X	3	Average of 3 determinations to nearest 1000 p.s.i.	2 foot length of each (as applicable)

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TABLE III.- Test methods of components (cont'd)

Component and lot expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	Number determinations per sample unit	Results reported as	Sample unit
Coil (cont'd)	Diameter	3.1.11	Standard Commercial	X	-	1	Pass or fail	2 foot length of each (as applicable)
	Protective coating	3.1.11.1	4.3.3.2	X	-	1	Pass or fail	
Tufting buttons (gross)	Protective coating	3.1.14	4.3.3.2	X	-	1	Pass or fail	1 dozen
Helical wire (foot)	Material identification	3.1.12	Standard commercial	X	-	1	Pass or fail	
	Diameter	3.1.12	Standard commercial	X	-	1	Pass or fail	2 footlength of each (as applicable)
Helical (foot)	Diameter	3.3.2.1	Standard commercial	X	-	1	Pass or fail	2 foot length of each
Stabilizer wire (foot)	Diameter	3.1.15	Standard commercial	X	-	1	Pass or fail	2 foot length
Steel wire, for wire mat of type II mattresses (1 coil)	Material identification	3.1.16	Standard commercial	X	-	1	Pass or fail	2 foot length
	Tensile strength	3.1.16	Standard commercial	-	X	3	Average of 3 determinations to nearest 1000 p.s.i.	2 foot length
	Diameter	3.1.16	Standard commercial					

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TABLE I.- Test methods of components (cont'd)

Component and lot expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	Number determinations per sample unit	Results reported as	Sample unit
Fibre cord, for wire mat of type II mattresses	Material identification	3.1.16	Standard commercial	X	-	1	Pass or fail	
Center cord	Breaking strength	3.1.16	4102, CCC-T-191	X	-	-	-	1 spool or cone
	Diameter	3.1.16	Standard commercial					
Edge cord (1 spool or cone)	Breaking strength	3.1.16	4102, CCC-T-191	X	-	-	-	1 spool or cone
	Diameter	3.1.16	Standard commercial					
Coils, each type (100 coils)	Corrosion resistance	3.1.11.1	4.3.3.2	X	-	1	Pass or fail	1 coil
Coils (type I mattress)	Compression set	3.1.11.1.2	4.3.3.1	X	-	1	Pass or fail	2 coils
Coils (type II & III mattresses)	Compression set	3.1.11.2.2 or 3.1.11.3.2 (as applicable)	4.3.3.1	X	-	1	Pass or fail	2 coils of each type (as applicable)

Note: Samples of cotton staple and linters shall not be sent to the Government laboratory for grading if the cotton has been officially graded by the United States Department of Agriculture. The inspector shall obtain from the contractor a certificate or an attested or photostatic copy of the certificate showing the number of bales, the number or identifying mark on each bale graded, and the identity of the grade when United States Department of Agriculture graded contractor-supplied material is used. Whenever contractor-furnished cotton staple or linters have not been graded by a properly qualified employee of the Government or a Government licensed classifier, the sample of cotton staple or linters or both shall be sent to the Government laboratory for grading. Grading by any other than Government employed graders shall not be acceptable.

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4.2.3.3 *Examination of the all-wire spring unit, knotted-coil-construction assembly (type III mattresses) for visual defects.* The sample unit for this examination shall be one type III mattress. Examination of the spring-unit assembly for defects in material, design, and workmanship:

Examine	Defect
Construction	Spring unit does not consist of either transverse rows of coils connected by the use of helicals running at right angles to the long axis of the unit as specified; or the spring unit does not contain transverse rows of individual coils, or units of not more than 4 coils to a unit joined to the neighboring coils or units as specified.
	Coil contains less than three active convolutions or turns (including only portions between knots of the coil).
	Coil height is less than 4-1/2 inches or more than 5 inches.
	Outside diameter of helicals less than 1/4 inch or more than 3/8 inch.
	Helicals contain less than two turns per inch.
	Note: When helicals are used in the construction, the individual spring coils may be offset with a portion or portions passing through not less than three turns of the helical.
	All knots not turned down.
	Any end of any helical not turned in so as to form a smooth and rounded termination of the helicals.

4.2.3.4 *Examination of spring unit assemblies (types I, II, and III mattresses) for number of coils in each unit.* Any number of coils per row (in direction of length or width) less than specified shall be classified as a defect.

4.2.3.5 *Inspection levels and acceptable quality levels (AQL's).* Inspection levels and AQL's expressed in defects per hundred units shall be as follows:

For examination in	Inspection levels	AQL's
4.2.3.1	L 6	one class 4.0
4.2.3.2	L 6	one class 4.0
4.2.3.3	L 6	one class 4.0
4.2.3.4	L 6	one class 6.5

4.2.4 *Inspection of the end item.*

4.2.4.1 *Visual examination of mattresses.* The defects found during examination shall be classified in accordance with 4.2.4.1.1, 4.2.4.1.2, and 4.2.4.1.3. The lots shall be inspected in accordance with 4.2.4.1.4. The sample unit for these examinations shall be one mattress (as applicable).

4.2.4.1.1 *Examination of mattress for defects in design, material, construction and workmanship.*

Examine	Defect	Classification	
		Major	Minor
Quality of ticking	Any hole, cut, or tear (except underneath tuft)	X	
	Any weak place seriously affecting serviceability	X	
	Any weak place affecting serviceability but not seriously		X
	Any smash	X	
	Rancid or other objectionable odor	X	
	Not fabric specified	X	
	Multiple floats or skips 1/2 inch or more in combined directions	X	
	Multiple floats or skips less than 1/2 inch in combined directions		X
	Three or more contiguous broken or missing warp or filling threads	X	
	Two contiguous broken or missing warp or filling threads		X

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Examine	Defect	Classification	
		Major	Minor
Quality of ticking (cont'd)	Loose and weakening slub, 1/8 inch or more in thickness	X	
	Spot or stain 1/4 inch or over in diameter, clearly noticeable when viewed at a distance of 6 feet		X
Seams and stitching	Any required seam or stitching operation omitted	X	
	Ticking not seamed with seam type SSa-1 or LSb-2		X
	Not stitch type 301 or 401		X
	Five—seven stitches per inch		X
	Less than five stitches per inch	X	
	More than twelve stitches per inch		X
	Ends of any seam not backstitched (except when end of seam is caught in another seam or stitching)		X
	Raw edge more than 1 inch in length		X
	Open seam more than 1 inch in length	X	
	Open seam 1 inch or less in length		X
	Note: A seam shall be classified as open when one or more stitches are broken, or when two or more continuous skipped stitches or run-offs occur.		
	Stitch tension loose, resulting in a loosely secured seam		X
	Stitch tension tight (stitches break when normal strain is applied to the seam or stitching)		X
Seams and stitching (cont'd)	Note: Puckering is evidence of tight tension, when puckering is evident, the inspector shall test seams by exerting pull on seam.		
	Needle chew which may result in hole or cut	X	
	Seam badly puckered, pleated, twisted, or caught in unrelated operation		X
	Untrimmed thread ends throughout		X
Construction	Not type specified	X	
	Sagging appearance, i.e., filling not evenly distributed	X	
	Lumps clearly noticeable	X	
	Thin hollow places clearly noticeable	X	
	Border or corner contains insufficient padding		X
	Spring coils can be felt through ticking	X	
	Face-panel ticking more than two pieces of material	X	
	Blue strips of ticking does not run in lengthwise direction		X
	Ticking stripes not properly matched adversely affecting appearance		X
Border and bound edges	Border not constructed as specified	X	
	Border not fabricated of one piece of ticking (all around)		X
	Transverse stitching on border does not extend across the full width of the border		X

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Examine	Defect	Classification	
		Major	Minor
Border and bound edges (cont'd)	Transverse stitching on border less than 1-1/4 inch on centers or more than 1-3/4 inch on centers		X
	Not box edge or box edge not sharply defined		X
	Binding tape does not extend completely around edge of mattress		X
	Binding tape pleated or twisted to an extent affecting serviceability	X	
	Binding tape puckered, pleated, or twisted affecting appearance but not serviceability		X
	Ventilators not approved type		X
	Ventilator insecurely attached or not firmly anchored to the ticking		X
	Less than four ventilators in each side	X	
	Less than two ventilators in each end of sizes 1 through 6 and size 13 mattresses	X	
	Less than three ventilators in each end of sizes 7 through 12 mattresses	X	
	Evidence that ventilator passage is obstructed to an extent interfering with free passage of air into the spring section		X
	Any ventilator not aligned halfway between the top and bottom of border (permit ± 1/4 inch tolerance)		X
	Ventilators not equidistant on either side of ends of mattresses (permit ± 1 inch tolerance)		X
	Ventilator screen damaged		X
	Less than two lifting straps on each side of mattress	X	
	Any strap misplaced, misaligned or too short affecting appearance		X
	Vertical strap (when applicable) not fabricated of double thickness ticking		X
	Vertical strap (when applicable) not positioned at 1/3 distance from each end (permit ± 2 inches tolerance)		X
	Sewn edges of vertical strap (when applicable) not turned in		X
	Vertical strap (when applicable) attached to border with less than four rows of stitching		X
	Stitching used to attach vertical strap (when applicable) to border crisscrossed or is more than 5/16 or less than 3/16 inch apart	X	
	Vertical strap (when applicable) not caught under the tape binding		X
	Horizontal lifting-handle pulls out of retainer (check by lifting mattress, using all handles)	X	
	Horizontal lifting-handle grommet or eyelet insecurely attached to the ticking		X

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Examine	Defect	Classification	
		Major	Minor
Tufting	Not type specified	X	
	Not accomplished as specified		X
	Tufts do not appear on both sides of mattress	X	
	Depth of button less than 3/4 inch from surface of mattress (check a minimum of three buttons on each side)		X
	Less than the number of tufts specified in table I		X
	More than one button or tape tied tuft or tuft with locking strip loose or insecure		X
	Knots of tape tying tufts (when applicable) appear above surface on either side of mattress		X
	Tear over 3/4 inch in ticking under button (check underneath a minimum of three buttons on each side of mattress)		X
	Arrangement of diamond pattern not as specified		X
	Spacing of button tufts not uniform affecting appearance		X
Labels	Missing, incomplete, incorrect, or illegible		X

4.2.4.1.2 *Examination of mattress for defects in dimensions.*

Examine	Defect
Thickness	Average is less than 6-1/4 inch. Average is more than 6-3/4 inch.
Length	Average length is beyond tolerances indicated in table II.
Width	Average width is beyond tolerances indicated in table II
Diameter of tufting button	Less than 1/2 inch. More than 5/8 inch.
Radius at corners of mattress	Less than 4 inches. More than 5 inches.
Width of vertical lifting straps (when applicable)	Less than 1-5/8 inches. More than 1-7/8 inches.
Diameter of horizontal lifting handle (when applicable)	Less than 5/16 inch. More than 7/16 inch.
Distance of outer tufted row to edge of mattress	Less than 5 inches. More than 7 inches.

4.2.4.1.3 *Examination of mattress for defects in weight.*

Examine	Defect
Weight (as applicable)	Less than minimum weight specified in table II More than maximum weight specified in table II

4.2.4.1.4 *Inspection levels and acceptable quality levels (AQL's).* Inspection levels and AQL's expressed in defects per hundred units shall be as follows:

For examination in	Inspection level	AQL
4.2.4.1.1	II	Major 4.0 Total 10.0
4.2.4.1.2	L 6	1 class 4.0
4.2.4.1.3	L 6	1 class 4.0

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TABLE IV.- Test methods for mattresses

Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	Number determinations per sample unit	Results reported as	Inspection level	AQL
Deviation of mattress surface (type I mattress)	3.7	4.3.1	-	X	10	To nearest degree	L 2	*
Depression under compression plunger (type I mattress)	3.8	4.3.2	-	X	15 (5 in each of 3 parallel lines)	To nearest 1.0 percent	L 2	-
Folding of mattress (type III mattress)	3.3.3.1	Fold or roll mattress from end to end	X	-	1	Pass or fail	L 1	4.0

4.2.4.2 *Testing of the end item.* Testing of the completely fabricated mattress shall be performed in accordance with table IV for the characteristics shown therein. The sample unit shall be one mattress, completely fabricated. The inspection level (sample size) shall be determined from MIL-STD-105.

4.2.5 *Examination of preparation for delivery requirements.* An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be level L 4 and the AQL shall be 4.0 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Content	Number per container is more or less than required.

4.3 Test methods.

4.3.1 *Tripod-level test (type I mattress).* The uniformity of mattress compression shall be determined by the use of a swivel level mounted on a tripod base. The instrument shall be placed on the mattress and oriented at random at 10 or more places to indicate, in degrees, the maximum deviation of the mattress top from a horizontal plane. During the test no leg of the tripod shall be

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located directly on a tuft nor less than 1-1/2 inches from the edge of the mattress. The tripod level consists of a base plate supported by three 1-1/4-inch ball-pointed legs, 3 inches long and 6 inches center to center. Upon the base plate is mounted a 9-inch quadrant divided into forty-five 1-degree divisions, each side of a plumbline center. A plumb bob is suspended from the apex of the quadrant. The quadrant is swivel mounted in the center of the base plate. The weight of the tripod level complete is 10 pounds.

4.3.2 *Compression-plunger test across mattress (type I mattress).* This test consists in compressing the mattress with a ball-pointed plunger 1-1/4 inches in diameter using a 20-pound load on the plunger. The depth of depression is noted at intervals of 1-1/4 inches in three parallel lines 1-1/4 inches apart across the mattress in right-angle directions.

4.3.3 *Coils.*

4.3.3.1 *Coil compression.* Coil compression shall be accomplished by placing either end of a single coil on a flat surface and then compressing it manually, using a board or block over the top of the coil, until all adjacent convolutions are touching or are compressed to a height of not more than 2 inches. This procedure shall be repeated 20 times to determine conformity with the requirements specified in 3.1.11.1, 3.1.11.2, and 3.1.11.3. If a large number of coils must be tested, a simple mechanical reciprocating type testing device may be used to accomplish the above test.

4.3.3.2 *Salt-spray test.* The sample coils for type I mattresses and metal tufting buttons furnished shall be subjected to a salt-spray test consisting of exposure in a closed tank of neutral material, such as wood or soapstone, to a finely divided spray at room temperature (65° to 80°F.) of 4 or 5 percent sodium chloride (by weight) in distilled water for a period of 72 hours. The tank shall be designed so that there will be no direct impingement of the spray upon the test specimens, and the spray shall circulate freely about all specimens to the same degree. None of the test specimens shall show signs of failure of protective coating, that is, the appearance of rust on the steel.

4.3.4 *Tufting twine, braid, or tape and fiber cord.*

4.3.4.1 *Breaking strength.* The sample tufting twine or braid and fiber cord shall be tested for breaking strength in accordance with method 4102 of CCC-T-191; tufting tape in accordance with method 5100, except that the full width of the tape shall be used.

5. PREPARATION FOR DELIVERY

Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.

5.1 **Packaging.** Packaging shall be level A or C, as specified (6.2).

5.1.1 *Level A.* Each mattress shall be individually packaged in a snug-fitting polyethylene bag. The polyethylene flexible film shall be 0.004-inch thick conforming to type I, grade A, B, or C, finish 1 of L-P-378. All seams and closures shall be effected by heat sealing and shall conform to the heat-seal strength requirement in L-P-378. Prior to or during the final heat sealing operation excess air within the bag shall be expelled.

5.1.2 *Level C.* Mattresses shall be packaged in accordance with the industry's practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (6.2).

5.2.1 *Level A.*

5.2.1.1 *Bulk packing.* Three mattresses

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of one type and size only, packaged as specified in 5.1, shall be packed flat in a snug-fitting shipping container conforming to overseas type, style A of PPP-B-576 except for the following modifications: The top and bottom shall have five equally spaced cleats (edge and intermediate cleats), extending the full width of the panel, and edge filler cleats shall be applied between the width cleats. Thickness of paper-overlaid veneer board shall be not less than 0.1570 inch thick and cleats shall be not less than 1-3/4 inches in width by 3/4 inch thick. Alternatively, shipping containers may be used conforming to overseas type, style A, type III, class 1 of PPP-B-601, except that the plywood shall be of 3/16-inch thickness, and cleats and arrangement shall be the same as specified above: or class 2, grade A, style E of PPP-B-640. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container specification, except when fiberboard shipping containers are to be closed with steel strapping, that only three straps be applied girthwise.

5.2.1.2 *Individual packing* (6.2). When specified, each mattress of one type and size only, packaged as specified in 5.1, shall be packed flat in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s, style FOL of PPP-B-636. Each shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. Shipping container dimensional and weight requirements will be waived.

5.2.2 *Level B.*

5.2.2.1 *Bulk packing.* Three mattresses of one type and size only, packaged as specified in 5.1, shall be packed flat in a snug-fitting shipping container conforming to domestic type, style A of PPP-B-576 except for the following modification: The top and bottom of the box shall have five equally spaced cleats (edge and intermediate cleats), extending the full width of the panel, and edge filler cleats shall be applied between the width cleats. Thickness of paper-overlaid veneer board shall be not less than 0.1570 inch thick and cleats shall be not less than 1-3/4 inches in width by 3/4 inch thick. Alternatively, shipping containers may be used conforming to domestic type, style A, type III, class 1 of PPP-B-601, except that the cleats and arrangement shall be the same as specified above.

5.2.2.2 *Individual packing* (6.2). When specified, each mattress of one type and size only, packaged as specified in 5.1, shall be packed flat in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style FOL of PPP-B-636. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps, and sealing the top flaps with adhesive. Shipping container dimensional requirements will be waived.

5.2.3 *Level C.* Mattresses, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129. The following marking shall appear in capi-

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tal letters not less than 1-1/2 inches in height on at least two sides of each shipping container:

"STOW FLAT — DO NOT STOW ON END"

5.3.2.1 *Polyethylene-bagged packages.* Polyethylene-bagged packages shall have the required information legibly printed or stamped in black on a white paper label inserted within the bag at one end of the mattress so as to permit ready identification. The label shall indicate the following information:

STOCK NUMBER
NOMENCLATURE
TYPE
SIZES
QUANTITY
CONTRACT NUMBER

6. NOTES

6.1 Intended use. These mattresses are intended for use by civil and military agencies of the Government.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and size of mattress required (1.2 and 1.2.1).
(*c*) Whether medical label is required (3.10.3).
(*d*) Whether alternate lifting straps are desired (3.6).
(*e*) Selection of applicable levels of packaging and packing (5.1 and 5.2).

6.3 Transportation description. The transportation description applicable to this item is:

Carload minimum weights:
In cars 40 feet 9 inches in length, inside measurement; 10,000; 12,000; 15,000; 18,000; 22,000; 30,000 pounds.
In cars exceeding 40 feet 9 inches in length, inside measurement; 16,000; 18,000; 21,000; 24,000; 30,000; 40,000 pounds.
Truckload minimum weight 5,600 pounds.

6.4 Standard sample. For access to a standard sample of felt address the procuring office issuing the invitation for bids.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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